# MAT2410 - Introduction to complex analysis: Mandatory assignment 2 of 2

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Autumn 2026

## Problem 1

Let $f(z) = \frac{e^z}{(1-z)^2}$, and let $\sum_{k=0}^{\infty} a_k z^k$ be the power series expansion of $f$ around 0.

**(a)** Determine $a_0$, $a_1$ and $a_2$.

*Proof.* Using the formula $a_k = \frac{f^{(k)}(0)}{k!}$, we have:

$$\begin{aligned} a_0 &= f(0) = \left.\frac{e^z}{(1-z)^2}\right|_{z=0} = 1 \\ a_1 &= f'(0) = \left.\frac{e^z(3-z)}{(1-z)^3}\right|_{z=0} = 3 \\ a_2 &= \frac{f''(0)}{2} = \frac{1}{2}\left.\frac{e^z(z^2-6z+11)}{(1-z)^4}\right|_{z=0} = \frac{11}{2} \end{aligned}$$

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**(b)** Determine the radius of convergence of the power series.

*Proof.* The radius of convergence of the power series at $z = 0$ is equal to the distance to the nearest singularity of $f$. Since the only singularity is a pole at $z = 1$, the radius of convergence is 1. ■

**(c)** Compute $\int_{|z|=2} f(z)\,dz$, where the path around the circle $|z| = 2$ is oriented counterclockwise.

*Proof.* Let $g(z) = e^z$. By applying Cauchy's integral formula for derivatives we get:

$$e = g'(1) = \frac{1}{2\pi i}\oint_{|z|=2} \frac{g(z)}{(z-1)^2}\,dz = \frac{1}{2\pi i}\oint_{|z|=2} \frac{e^z}{(1-z)^2}\,dz.$$

Rearranging this equation, we get:

$$\oint_{|z|=2} f(z)\,dz = 2\pi i e.$$

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## Problem 2

Let $D \subseteq \mathbb{C}$ be a bounded domain with a piecewise smooth boundary. Let $f$ be analytic on $D$, extending smoothly to $\overline{D}$.

**(a)** Show that if $f$ is constant on $\partial D$, then $f$ is constant on $\overline{D}$.

*Proof.* Since $f$ is constant on $\partial D$, there exists a constant $c \in \mathbb{C}$ such that $f(z) = c$ for all $z \in \partial D$. Let $z \in D$. By the Cauchy integral formula, we have:

$$f(z) = \frac{1}{2\pi i} \oint_{\partial D} \frac{f(w)}{w - z}\, dw = \frac{c}{2\pi i} \oint_{\partial D} \frac{1}{w - z}\, dw = c$$

Thus, $f(z) = c$ for all $z \in D$. Since $f$ equals the same constant on both $D$ and $\partial D$, it is constant on $\overline{D}$. ■

**(b)** Show that if $f(z) \neq 0$ for all $z \in \partial D$ and

$$\int_{\partial D} \frac{1}{f(z)}\, dz \neq 0,$$

then $f$ has a zero in $D$.

*Proof.* We prove the contrapositive. Assume that $f$ has no zeros in $D$. Since $f$ is also non-zero on $\partial D$, the function $\frac{1}{f(z)}$ is analytic on $D$ and extends smoothly to $\overline{D}$. By Cauchy's integral theorem, we get:

$$\int_{\partial D} \frac{1}{f(z)}\, dz = 0.$$

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## Problem 3

Let $\alpha \in \mathbb{C}$. For each integer $k \geq 1$, let

$$\binom{\alpha}{k} = \frac{\alpha(\alpha - 1) \ldots (\alpha - k + 1)}{k!},$$

and let $\binom{\alpha}{0} = 1$. Consider the power series

$$f(z) = \sum_{k=0}^{\infty} \binom{\alpha}{k} z^k.$$

**(a)** Determine the radius of convergence of the power series; the answer will depend on whether $\alpha \in \mathbb{Z}_{\geq 0}$ or not.

*Proof.* If $\alpha \in \mathbb{Z}_{\geq 0}$, we have $\binom{\alpha}{k} = 0$ for all $k > \alpha$. In this case, the power series terminates and reduces to a polynomial, which has a radius of convergence of $\infty$.

Otherwise, if $\alpha \in \mathbb{C} \setminus \mathbb{Z}_{\geq 0}$, then $\binom{\alpha}{k} \neq 0$ for all $k \geq 0$. Applying the ratio test, we get:

$$\lim_{k \to \infty} \left| \frac{\binom{\alpha}{k}}{\binom{\alpha}{k+1}} \right| = \lim_{k \to \infty} \left| \frac{k+1}{\alpha - k} \right| = 1.$$

Therefore, the radius of convergence is 1 for $\alpha \notin \mathbb{Z}_{\geq 0}$. ■

**(b)** Recall that for $w \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$ we define the principal branch $P_\alpha$ of the $\alpha$-th power function $w^\alpha$ by

$$P_\alpha(w) = e^{\alpha \operatorname{Log}(w)},$$

where $\operatorname{Log}(w)$ denotes the principal branch of the logarithm. Show that

$$P'_\alpha(w) = \alpha P_{\alpha-1}(w)$$

for all $w \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$.

*Proof.* Let $w \in \mathbb{C} \setminus \mathbb{R}_{\leq 0}$. Applying the chain rule, we have:

$$\begin{aligned} P'_\alpha(w) &= \frac{d}{dw} e^{\alpha \operatorname{Log}(w)} \\ &= e^{\alpha \operatorname{Log}(w)} \cdot \frac{\alpha}{w} \\ &= \alpha e^{\alpha \operatorname{Log}(w)} e^{-\operatorname{Log}(w)} \\ &= \alpha e^{(\alpha-1)\operatorname{Log}(w)} \\ &= \alpha P_{\alpha-1}(w). \end{aligned}$$

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**(c)** Determine $P_\alpha^{(k)}(1)$ for all $k \geq 0$ and use this to show that for $|z| < 1$, we have

$$f(z) = P_\alpha(1+z).$$

*Proof.* By the recursive formula derived in the previous part, it follows that $P_\alpha^{(k)}(1) = \alpha(\alpha - 1)\cdots(\alpha - k + 1)$ for $k \geq 1$, and define $P_\alpha^{(0)}(1) = 1$. Thus, the power series expansion of $P_\alpha(z+1)$ around $z = 0$ for $|z| < 1$ gives:

$$P_\alpha(z+1) = \sum_{k=0}^{\infty} \frac{P_\alpha^{(k)}(1)}{k!} z^k = \sum_{k=0}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!} z^k = \sum_{k=0}^{\infty} \binom{\alpha}{k} z^k = f(z).$$

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## Problem 4

Let $f(z) = \frac{e^z}{z}$, let $r > 0$, and let $\gamma_r$ be the path

$$\gamma_r(t) = -r\sin(t) + ir\cos(t), \quad 0 \leq t \leq \pi.$$

Show that

$$\left| \int_{\gamma_r} f(z)\, dz \right| \leq \pi.$$

*Proof.* The length of the path is $L := \pi r$. Let $z$ be in the image of $\gamma_r$. We have:

$$|f(z)| = \left| \frac{e^z}{z} \right| = \frac{|e^z|}{|z|} = \frac{e^{\operatorname{Re}(z)}}{r} \leq \frac{e^0}{r} = \frac{1}{r} =: M.$$

By the ML-estimate, it follows that:

$$\left| \int_{\gamma_r} f(z)\, dz \right| \leq ML = \pi.$$

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## Problem 5

For each of the hypothetical function descriptions below, either give an example of a function satisfying the description, or prove that no such function can exist.

**(a)** A function which is defined and analytic on $\mathbb{C} \setminus \mathbb{Z}$, but does not admit an analytic extension to any point of $\mathbb{Z}$.

*Proof.* Let $f(z) = \frac{1}{\sin(\pi z)}$. This function has non-removable poles exactly at the integers, and therefore cannot be analytically extended to any point in $\mathbb{Z}$. ■

**(b)** An entire function such that $|f^{(k)}(0)| \geq k!$ for all $k \geq 0$.

*Proof.* Suppose for the sake of contradiction that there exists such a function. Consider its power series expansion around $z = 0$. Since $f$ is entire, the radius of convergence is $\infty$:

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k$$

This series must converge absolutely for all $z \in \mathbb{C}$, and in particular for $z = 1$. However, we have:

$$\sum_{k=0}^{\infty} \left| \frac{f^{(k)}(0)}{k!} 1^k \right| \geq \sum_{k=0}^{\infty} \frac{k!}{k!} = \sum_{k=0}^{\infty} 1 = \infty,$$

which diverges. This is a contradiction. ■

**(c)** An entire function such that $f\left(\frac{1}{n}\right) = \frac{1}{2-1/n}$ for all integers $n \geq 1$.

*Proof.* Suppose for contradiction that there exists such an entire function $f$. Define $g(z) = \frac{1}{2-z}$. Then $g\left(\frac{1}{n}\right) = \frac{1}{2-1/n}$ for all $n \geq 1$. Since $f$ is continuous, we have:

$$f(0) = \lim_{n \to \infty} f\left(\frac{1}{n}\right) = \frac{1}{2} = g(0).$$

Since every neighborhood of 0 contains points of the sequence $(1/n)$, the set of points where $f$ and $g$ agree contains a non-isolated point at $z = 0$. By the Uniqueness Principle $f = g$. However, this is a contradiction since $g$ has a pole at $z = 2$ and therefore cannot be extended to $\mathbb{C}$. ■